Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

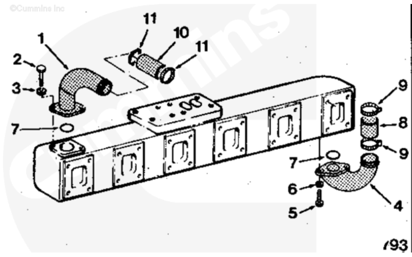
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html)
- Remove the water bypasss tube. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html)

Remove

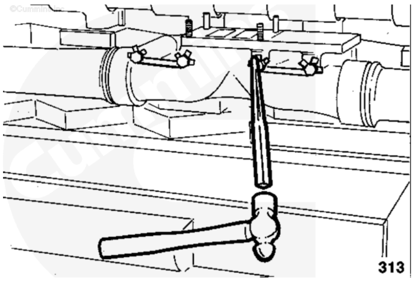
Remove the water transfer connections (1) and (4) from the wet exhaust manifold.

Remove and discard the two o-rings (7), and hoses (8) and (10).



Use a drift and a hammer to bend the lockplate tabs away from the capscrew heads.

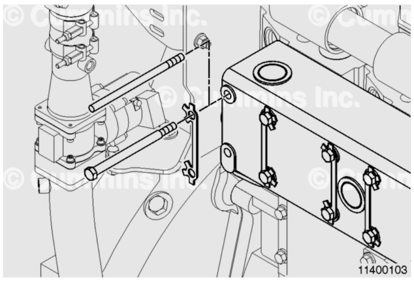
Note : Some engines do not use this type of lockplate. Some engines use a lockplate that snaps over the capscrew head and some engines use locking capscrews that do not require lockplates.



Remove the two exhaust manifold capscrews and lockplates.

Use two 7/16-14 x 15 inch guide studs to support the manifold during removal.

Install the guide studs.



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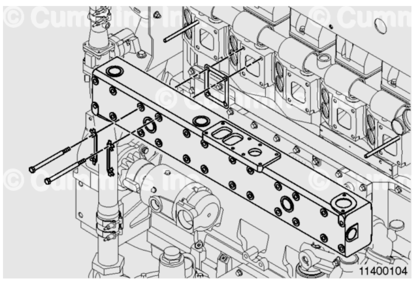


Remove the remaining capscrews and lockplates.

Remove the exhaust manifold and gaskets.

Discard the gaskets.

Clean and Inspect for Reuse

⚠ WARNING ⚠

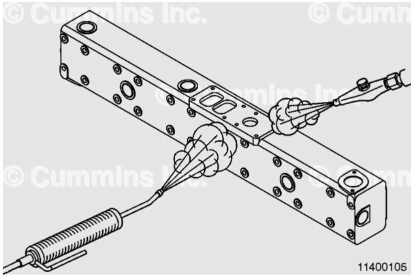
Wear safety glasses or a face shield, as well as protective clothing, to prevent personal injury when using a steam cleaner or high-pressure water.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

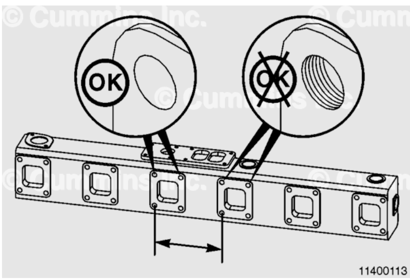
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use steam to clean the exhaust manifold sections.

Use a wire brush to remove any scale from the outside and inside diameters of the section sealing joints.

Inspect the manifold sections for cracks in the areas shown.



Inspect the capscrew holes in the center section for damage.

High temperature can cause the manifold to deform. If capscrew thread marks are visible on the sides of the capscrew holes, the manifold has been subjected to high temperature.

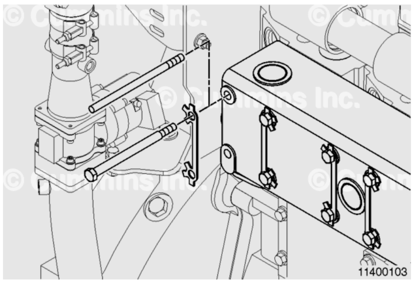
Measure the center-to-center distance between the same position capscrew holes of the two flanges on the center section and both end sections.

Exhaust Manifold Capscrew Holes		
mm		in
191.5	MIN	7.54
193.5	MAX	7.62

Install

⚠ CAUTION ⚠

Do not use gasket cement to hold the exhaust manifold gaskets in place during installation. The use of gasket cement lead to an exhaust gas leak.



Use two 7/16-14 x 5 inch guide studs to support the manifold during installation.

Install the exhaust manifold gaskets onto the cylinder head. Use a contact adhesive to hold the gaskets in place during installation.

Install the exhaust manifold onto the guide studs.

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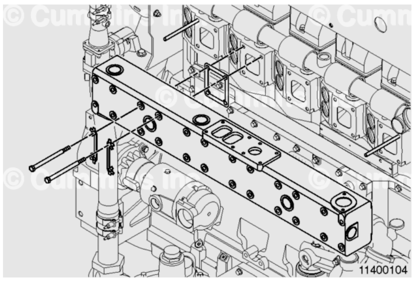


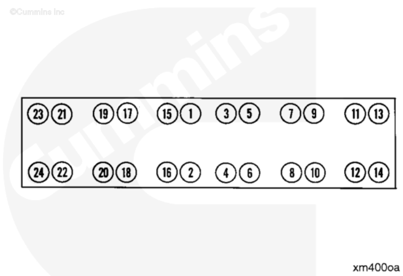
Apply anti-seize compound to the exhaust mounting capscrew threads.

Install the lockplates and capscrews.

Remove the guide studs (12) and install the remaining capscrews.

Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

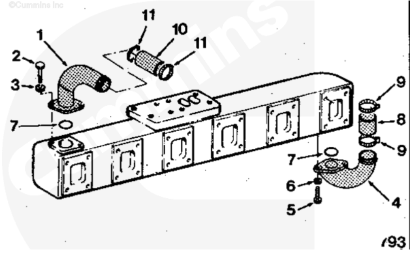


Slide the hose (10) and hose clamps (11) on the front water connection (1). Install the sealing ring (7) in the flange.

Install the front water connection, washers (3), and capscrews (2).

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Connect the front water connection hose to the transfer tube between cylinder number 1 and cylinder number 2.

Tighten the hose clamps.

Torque Value: 5.6 n•m [50 in-lb]

Slide the hose (8) and hose clamps (9) on the rear water connection (4). Install the sealing ring (7) in the flange.

Install the rear water connection, washers (6), and capscrews (5).

Tighten the capscrews.

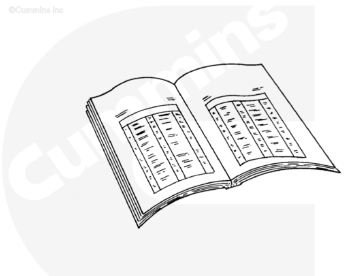
Torque Value: 45 n•m [33 ft-lb]

Connect the rear water connection hose to the transfer tube between cylinder number 5 and cylinder number 6.

Tighten the hose clamps.

Torque Value: 5.6 n•m [50 in-lb]

Finishing Steps



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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the water bypass tube. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html)
- Install the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-

018-tr.html)

- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for leaks.

Last Modified: 20-Aug-2015